

BioCoLoop: COLLABORATIVE AGENTIC RESEARCH FOR BIOLOGICAL MODEL IMPROVEMENT

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ABSTRACT

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access. A key challenge for self-improving AI in this setting is to use such distributed evidence to improve both model parameters and the computational solutions built around them. Collaborative learning typically optimizes fixed models, whereas automated model design and research generally rely on centralized training and evaluation signals. Bridging the two is nontrivial: distributed evidence must support not only optimization, but also the evaluation and refinement of computational changes that generalize beyond their local settings. To tackle this, we introduce BioCoLoop, a collaborative research framework that couples client-local parameter fitting with an evidence-guided proposal–evaluation loop. Local training shards update a shared predictor, while local development shards produce aggregate evidence cards that guide the next executable proposal. Across three biological settings spanning drug–target interaction prediction, observed-response proteomics, and cell perturbation, the complete system leads four of five main-table endpoints. Under matched data, candidate and compute budgets, evidence-guided allocation yields four held-out wins, eight ties and no losses across twelve task–seed comparisons, including a 10.84-point mean Top-1 gain on double-gene perturbations. BioCoLoop offers a practical path toward self-improving biological AI under data-sharing constraints.

1. INTRODUCTION

Biological data are often distributed across institutions and data holders, where privacy and data-sharing constraints limit centralized access (Sheller et al., 2020; Bakhtiari et al., 2025). Combining these measurements could improve models of molecular interactions, protein responses and cellular perturbations. Exploiting new measurements, however, involves more than refitting parameters: researchers also choose prediction heads, adjust optimization settings and decide which candidate models deserve further training. Automating this repeated process of proposing, testing and revising computational designs motivates self-improving biological AI. The challenge is to support that process when each laboratory retains its own data.

Figure 1 contrasts two complementary approaches. A centralized research agent iterates on data accessible to its execution environment, using experiment outcomes to revise candidate solutions. AIDE and the AI Scientist automate this process for computational research (Jiang et al., 2025b; Lu et al., 2024); DrugEvolve applies algorithm revision, evaluation and memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These workflows turn measurements into design feedback. Collaborative learning instead connects several data holders through local training and model-update aggregation (McMahan et al., 2017; Sheller et al., 2020; Wang et al., 2024). Here, a *client* is a laboratory’s computational worker, which operates on data held by that laboratory. This approach broadens the evidence available for fitting a specified model. Connecting the two requires a second shared interface: laboratories must also evaluate candidate designs and supply feedback that the research agent can use to choose its next experiment.

BioCoLoop couples these interfaces in a two-level research cycle. For any candidate design, an inner training process fits a shared predictor using laboratory-local data. An outer research loop uses the resulting development evidence to propose and select subsequent designs. In our implementation, a design specifies the trainable prediction head and its learning rate, regularization and aggregation set-

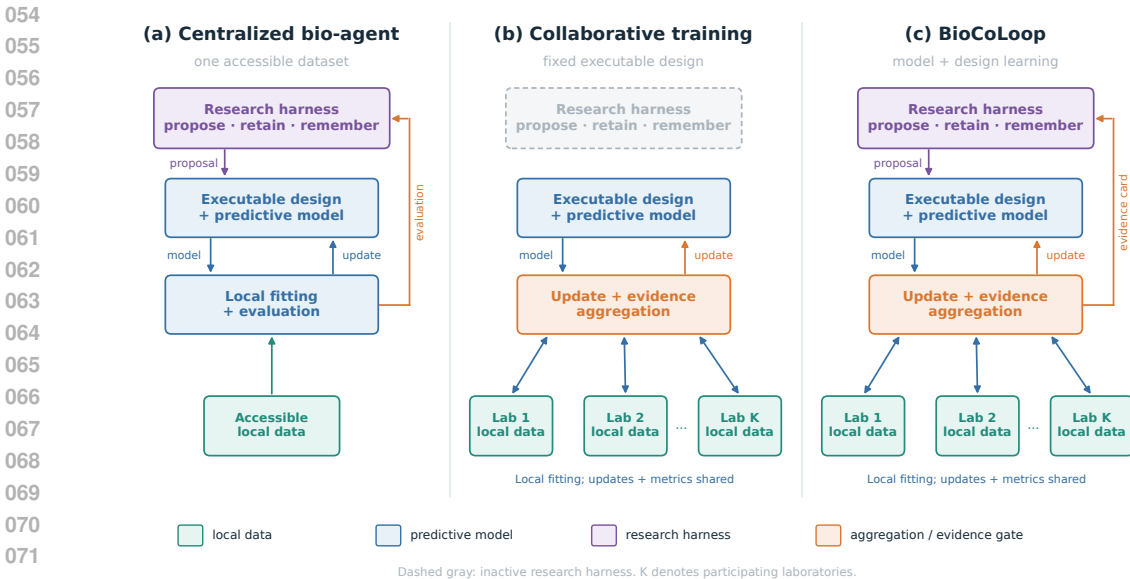


Figure 1: Three ways to use biological measurements. A centralized bio-agent revises a predictor using one accessible dataset. Collaborative training fits a specified model using data held by K laboratories. BioCoLoop also returns aggregate development evidence to the research loop, allowing distributed measurements to guide both parameter fitting and design revision. Colors identify data, models, research components and aggregation interfaces.

tings. The loop packages each hypothesis, executed configuration, learning trajectory and selection outcome into an *evidence card*. Accumulated cards condition the next proposal; early trajectories can also guide which candidates receive full training. Local measurements thus contribute both parameter updates and evidence for research decisions, while the language model receives aggregate diagnostics rather than raw biological responses.

We evaluate the framework in three biological settings: drug–target interaction (DTI) prediction, observed-response proteomic efficacy classification, and cell perturbation identification spanning single-gene, double-gene and drug interventions. The main comparison uses ten simulated laboratories for the collaborative system and one for its local references. Controlled comparisons then distinguish the effects of data access, design search and feedback-dependent allocation of training compute.

Our contributions are threefold. First, we connect laboratory-local parameter fitting to distributed evaluation of executable model designs through a common biological research interface. Second, we develop an evidence-guided loop mechanism that packages proposal outcomes into reusable research history and uses measured learning trajectories to select candidates for further validation. Third, we evaluate the shared mechanism across three task families, with matched-budget comparisons and sensitivity studies of laboratory participation, data sources and proposal budget. The contribution is this coupling and its experimental evaluation, with task-specific predictors connected through the same training and research interfaces.

2. RELATED WORK

Collaborative learning for biological data. Parameter aggregation enables institutions to train a shared model while retaining their measurements locally (McMahan et al., 2017; Sheller et al., 2020). In single-cell analysis, scFed addresses cell-type classification and FedscGen addresses distributed batch correction (Wang et al., 2024; Bakhtiari et al., 2025). Tabula and its Chiron implementation support collaborative single-cell foundation-model training through an agent-enabled platform (Ding et al., 2026; Mechtel et al., 2026). Automated design can interact with local training: FedEx tunes hyperparameters while training shared weights (Khodak et al., 2021), and Helmsman synthesizes

collaborative-learning systems through planning, coding and simulation-based refinement (Li et al., 2025). BioCoLoop connects parameter fitting to a second laboratory interface for evaluating proposed designs. Aggregate development evidence supports both proposal revision and allocation of further training.

AI for AI in biological research. Research agents turn experiment outcomes into subsequent computational proposals. AIDE searches candidate programs through drafting, debugging and improvement (Jiang et al., 2025b), while the AI Scientist connects ideas to executed experiments, replanning and writing (Lu et al., 2024). DrugEvolve applies research, implementation, evaluation and algorithm memory to drug-development tasks (Zhou et al., 2026; GAIR-NLP DrugEvolve contributors, 2026). These systems motivate executable revision and persistent experiment history. Our study asks how such feedback can be supplied by multiple data-owning laboratories through a shared research interface. We instantiate that interface with TAPB for DTI, a ProteinTalks-derived efficacy head for proteomic responses, and scDEBART-based cell representations (Lin et al., 2025; Sun et al., 2026; Sung & Kim, 2026). These models serve as task-specific prediction components; the proposal schema, local training service and evidence-driven selection mechanism remain common across tasks. The strong performance of simple references on some perturbation benchmarks further motivates separating the effect of a research loop from the choice of predictor (Ahlmann-Eltze et al., 2025).

3. METHOD: COLLABORATIVE AGENTIC BIOLOGICAL RESEARCH

3.1 FRAMEWORK OVERVIEW

BioCoLoop takes a task specification, an initial predictor, an editable design space and data held by K participating laboratories. The task specification defines prediction inputs and outputs, scorer and computational budget. Each laboratory maintains training and development partitions. A research controller proposes a design, a shared training service fits its parameters across laboratories, and local evaluators return development measurements to a coordinator. The outputs are the selected design with fitted parameters and an ordered history of proposals and outcomes (Figure 2).

The key mechanism is the return of experimental evidence to the research process. An *evidence card* links a proposed change to its executed configuration, aggregate learning trajectory and selection outcome. Successive cards form a persistent history, allowing the controller to revise its next hypothesis using both successful and unsuccessful trials. Trajectory information also guides training allocation: a candidate with promising early progress can be promoted for full validation. The outer loop decides what design to investigate and how to allocate evaluation; the inner process fits parameters for that design.

3.2 EVIDENCE-GUIDED PROPOSAL AND REVISION

The controller receives the task specification, current design and accumulated evidence cards. It returns a structured proposal with four fields: a *hypothesis* stating why a change should help, an *experiment* describing the comparison, an *expected_effect* identifying the predicted metric or diagnostic response, and a *design_id* resolving to an executable configuration. A validator checks the identifier against the task’s editable design space. The configuration determines the intervention; the other fields state its rationale and anticipated outcome. Appendix A gives the JSON schema and design library.

Local evaluators return scores, losses and learning-curve diagnostics. The coordinator packages these with the proposal, selected checkpoint and acceptance decision. It retains a candidate if its mean development score exceeds the incumbent’s, using lower mean loss to resolve ties. Each subsequent proposal receives the complete evidence history, including rejected directions. The research language model remains fixed; adaptation occurs in the predictive model and the designs explored by the loop.

Evidence influences two research decisions. In *proposal revision*, measured history conditions the next design. In *training allocation*, short learning trajectories determine which designs receive full validation. The latter uses a screening–promotion–validation policy that retains candidates with strong current scores as well as improving trajectories. These decisions share the evidence inter-

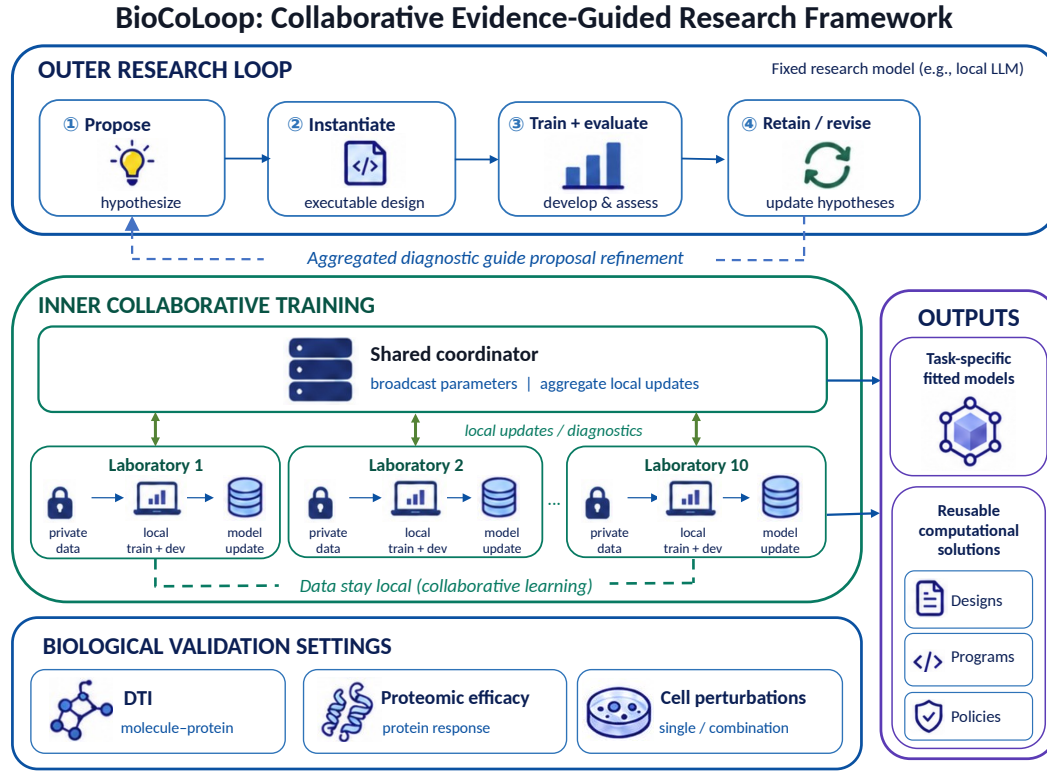


Figure 2: BioCoLoop couples an outer proposal–instantiation–evaluation–revision cycle with laboratory-local parameter fitting. The coordinator distributes parameters and aggregates local updates and development diagnostics; the fixed research model uses this feedback to revise designs. The illustrated ten-laboratory setting is the main experiment’s instance of a general K -laboratory interface. Reusable designs in the evaluated implementation specify prediction heads and optimization settings. Task adapters support DTI, observed-response proteomics and cell perturbation.

face and are evaluated separately: proposal-history comparisons assess design selection, while fixed-slate racing isolates evidence-dependent allocation. Section 4 and Appendix A specify budgets and promotion rules.

3.3 LOCAL FITTING AND DISTRIBUTED EVALUATION

For a selected design, laboratory workers start from shared parameters and fit them on local training partitions. The coordinator combines returned parameters using training-example weights, following FedAvg (McMahan et al., 2017), with optional server momentum specified by the design. The resulting predictor is distributed for the next round. This training pipeline is unchanged across task families; adapters define the input representation, prediction head and loss.

Development evaluation gives each laboratory equal weight. If $s_i(a)$ is laboratory i ’s development score for design a , the selection score is $S(a) = K^{-1} \sum_{i=1}^K s_i(a)$. Training-example weights determine parameter aggregation; equal-laboratory weights determine design and checkpoint selection. Every candidate starts from the same task-specific initialization for its seed. At completion, the selected checkpoint is fixed and evaluated by the held-out scorer.

Workers return parameter updates and example counts for training, and scalar scores and diagnostics for research feedback. The language model receives aggregate evidence cards. Raw interaction rows, proteomic profiles and expression responses remain at the worker interface. Experiments simulate these roles on one host, using $K = 10$ for the main comparison and varying K in the sensitivity study.

4. EXPERIMENTAL DESIGN

4.1 SHARED PROTOCOL AND COMPARISON METHODS

We evaluate whether collaborative data access and iterative research improve the same task-specific predictor. Training and development groups are assigned to ten disjoint simulated laboratories, balanced by compound or intervention identity. The single-laboratory setting uses preselected client 0; the collaborative setting uses all ten clients. Each comparison uses the same held-out scorer and candidate rosters.

Benchmark contracts.

Setting	Prediction contract	Task-reference model	Primary metric
DTI	Compound + protein $\rightarrow$ interaction	TAPB	AUROC $\uparrow$
Proteomics	Proteome + drug $\rightarrow$ efficacy	ProteinTalks-derived head	AP $\uparrow$
Cell (3 strata)	Response + 5 options $\rightarrow$ identity	Corrected scDEBART head	Macro Top-1 $\uparrow$

The cell contract is instantiated for VCC single-gene, Norman double-gene and Tahoe drug perturbations; task-specific held-out units and secondary metrics are specified below.

The main table compares three methods. The *task model* trains the reference architecture above with a fixed optimization recipe on one laboratory’s data. *Qwen direct* uses Qwen2.5-7B-Instruct to choose model configurations without receiving evaluation history; these configurations are trained on the same one laboratory. *BioCoLoop* uses the same proposer with evidence-card feedback and all ten laboratories. The language model proposes configurations; the task-specific predictor produces the biological predictions in every method. Direct and loop search share a twelve-design library, six proposal slots and the first proposal. Each candidate receives 100 training rounds, with development evaluation every five rounds. Appendix A records optimizer settings and the proposal interface.

Two controls separate the causes of improvement. A factorial comparison crosses one/ten laboratories with fixed/direct/loop research, using common seeds 42–44. It tests participation while holding research mode fixed, and research while holding participation fixed. A separate *compute-matched racing* comparison tests whether feedback improves training allocation: both methods receive the same ten designs and all ten laboratories. Uniform allocation gives each design 80 rounds. Evidence-guided allocation screens each for 20 rounds and restarts six promoted designs for 100-round validation. Both spend 800 full-client rounds and 160 development measurements. This experiment uses seeds 53–55 for proteomics and the three cell settings; its fixed candidate slate removes proposal generation as a varying factor.

Checkpoint selection uses participating laboratories’ development data, with equal weight per laboratory. Every selected predictor is evaluated on the common held-out test set. Means and sample SD summarize the specified training/search seeds on fixed partitions. Appendix A supplies all six configurations, secondary metrics and uncertainty estimates; bold marks all best displayed task scores, including ties.

4.2 DRUG–TARGET INTERACTION PREDICTION

DTI predicts a binary interaction from a compound and a protein sequence. We use the released BindingDB training/development split of 50,149/5,604 pairs. Canonical-compound groups are assigned to clients with a label-independent size-balancing rule, producing compound-disjoint clients while preserving the source split within each client.

The task reference uses the released TAPB architecture with frozen public ESM2 protein features and freshly initialized drug, fusion and classification modules (Lin et al., 2025). A target dictionary derived from client 0’s training proteins is shared identically by every arm, ensuring a common initialization. The objective combines TAPB’s classification NLL and masked-language loss under the shared batch-64 training pipeline. Candidate designs can add a zero-output residual adapter before TAPB’s native confounder-averaged prediction.

Selection uses mean-client AUROC; AP, MCC, accuracy at 0.5 and binary cross-entropy are secondary. The fixed test suite contains 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein

pairs. Random-split AUROC is the main-table endpoint, and both unseen-entity endpoints appear in the complete results. All six arms and three seeds complete the full protocol.

DrugBAN supplies an additional architecture reference under the uniform multi-laboratory trainer (Bai et al., 2023; DrugBAN contributors, 2026).

4.3 PROTEOMICS-BASED EFFICACY PREDICTION

We construct an observed-response efficacy benchmark from a subset of the public ProteinTalks PTPC supplement, which provides measured proteomic responses and binary efficacy labels (Sun et al., 2026). The task predicts efficacy from a treated condition’s observed 6h/24h proteome and drug representation. We partition the 491-condition source pool, spanning 333 compounds, into 386 fitting and 105 development conditions by compound group. Ten clients each own 49–50 total conditions, with compound-disjoint training and development groups and both classes in every local development set. Local fitting sets contain 35–42 conditions.

The task reference is a ProteinTalks-derived observed-response efficacy head: protein and drug convolutional branches followed by a 64–32–1 classifier, trained from random initialization on observed 6h/24h responses and Morgan features. A 24h linear/Morgan model provides a complementary non-neural reference.

Average precision (AP) is primary; AUROC, binary cross-entropy and accuracy at 0.5 are secondary. The 148-condition test role is evaluated after the development-selected checkpoint is frozen.

4.4 CELL PERTURBATION IDENTIFICATION

Given an observed expression response and five candidate interventions, the model selects the true perturbation by cosine agreement with predicted responses. We construct three inverse-identification benchmarks: VCC-derived single-gene perturbations from the public 2025 training release with internal intervention-disjoint partitions (Arc Institute, 2025); Norman double-gene overexpression (Norman et al., 2019); and a Tahoe-100M subset of 307 compounds in A549 at 5 μ M, from the February 25, 2025 release (Zhang et al., 2025). Whole-intervention train/development/test counts are respectively 60/20/20, 70/20/27 and 180/60/67. Each laboratory owns six/seven/eighteen training interventions and two/two/six development interventions. Assignment balances intervention identities, while retaining the naturally varying number of VCC cells per intervention.

The public CELLxGENE-pretrained scDEBART backbone is frozen and its response MLP is trainable (Sung & Kim, 2026). Three upstream attention masks are implemented as additive blocking masks and verified by response-free synthetic tests. This correction leaves pretrained weights, targets, splits and candidate rosters unchanged; features are then re-extracted for every arm.

Genetic tasks use the released perturbation encoding and exclude directly perturbed genes from loss and ranking coordinates. Tahoe adds fixed Morgan-projection conditioning for chemical identities. The uniform cell objective combines response MSE with 0.1 times a contrastive loss formed from local training conditions.

Primary scoring is intervention-macro Top-1, with a 20% random-choice expectation; micro Top-1, MRR, Top-3 and cosine-logit cross-entropy are secondary. All arms receive the same five candidate interventions for each response.

4.5 PARTICIPATION, INDEPENDENT SOURCES AND RESEARCH BUDGET

We vary laboratory count over $K \in \{1, 2, 5, 10\}$ for all five endpoints. Adding the first K original clients increases data access; merging all ten clients into K training groups keeps the total pool fixed. Both use the default recipe and 100 rounds, with the original development scoring units. DTI uses seed 61 and the four lighter endpoints use seeds 61–63.

Independent-source experiments add BioSNAP, Davis and Human to DTI, and Replogle, Nadig and Jiang to VCC. Target-only development selection and a common target test set make the effect of each source directly comparable. For proteomics, two independent studies supply measured protein responses and distinct efficacy endpoints (Lin et al., 2026; Ruprecht et al., 2020). A supplementary multi-task adapter shares protein/drug representations and retains source-specific output heads; its

target-only control uses the same architecture. A separate analysis retains three within-study mtPTDS contexts. Appendix B specifies source filtering, losses and all source combinations.

The proposer comparison uses Qwen2.5-7B-Instruct and gpt-5.6-luna across all five endpoints. Both explore the same twelve designs in six proposal slots, with test prefixes at 0, 1, 2, 4 and 6. A second, longer study explores 36 designs in 24 slots on proteomics and the three cell endpoints. Within each protocol, candidates receive 100 rounds from a common initialization, the first proposal is shared, and direct/loop differ in access to evaluation history. Failed slots consume the budget. DTI short-search results use seed 42; the remaining proposer studies use seed 61. The complete trajectories and fixed-prefix scores allow task-specific analysis of search progress and saturation.

5. RESULTS

5.1 MAIN HELD-OUT PERFORMANCE

Table 1 places the three task families in one comparison: DTI and proteomics each contribute one endpoint, and cell perturbation contributes three. Each cell reports mean $\pm$ sample SD over seeds 42–44. This SD measures training/search variability on fixed partitions. The overall column averages the five displayed endpoints within each seed before summarizing across seeds; it is a descriptive summary, not a common biological metric.

Method	DTI	Proteomics	Cell perturbation			Overall
	TAPB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$	Mean 5 endpoints
Task model (1 lab)	82.83 $\pm$ 1.22	<u>25.05</u> $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	19.40 $\pm$ 0.00	36.55 $\pm$ 0.82
Qwen direct (1 lab)	<u>89.89</u> $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21	32.14 $\pm$ 2.02
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59	41.39 $\pm$ 0.60

Table 1: Main held-out comparison. Task models are TAPB (DTI), a ProteinTalks-derived efficacy head (PTPC) and corrected scDEBART response heads (cells). Qwen direct proposes configurations without evaluation history; BioCoLoop adds collaborative access and evidence feedback. Values are mean $\pm$ sample SD, multiplied by 100; SD is not a confidence interval. Overall summarizes the five endpoints within each seed. Bold and underline mark the best and second-best displayed means. The shared training protocol is in Section 4; all six configurations, seed counts and secondary metrics are in Appendix A.

BioCoLoop leads four of five displayed endpoints. TAPB reaches 93.76% random-test AUROC versus 89.89% for one-laboratory direct optimization, and proteomic AP reaches 36.99% versus 20.13%. Relative to direct optimization, VCC, Norman and Tahoe macro Top-1 increase by +10.45, +8.10 and +6.97 points. Norman’s best displayed result remains the one-laboratory fixed recipe at 25.56%, compared with 22.25% for BioCoLoop. The complete system’s five-endpoint mean is 41.39, versus 36.55 for the fixed recipe and 32.14 for direct optimization. The main comparison measures collaborative access and research jointly; the following controls separate them.

5.2 DOES FEEDBACK IMPROVE THE ALLOCATION OF TRAINING COMPUTE?

This experiment asks whether early development measurements help allocate a fixed training budget. The candidate designs and laboratory data are identical. Uniform allocation trains every design equally; the evidence-guided loop screens all designs and promotes a subset using current performance and learning progress. Table 2 therefore measures training allocation with proposal generation held fixed.

Across 12 newly executed proteomics/cell task–seed comparisons, the loop wins four and ties eight. Norman improves from 11.48% to 22.32% macro Top-1, a +10.84-point mean effect with hierarchical paired-bootstrap 95% CI [4.54, 17.73]. Tahoe improves by +1.99 points, with one win and two ties; VCC and proteomics tie. On Norman, a promoted low-learning-rate residual design continues improving beyond the uniform method’s stopping point. Concentrating validation on promising candidates improves the final predictor at the same total training budget. The DTI development replay

Benchmark	Uniform	Evidence-guided	Gain	W/T/L
Proteomics: efficacy	37.47	37.47	+0.00	0/3/0
Cell: VCC single-gene	20.01	20.01	+0.00	0/3/0
Cell: Norman double-gene	11.48	22.32	+10.84	3/0/0
Cell: Tahoe drug	20.90	22.89	+1.99	1/2/0

Table 2: Does early evidence improve training allocation? Held-out scores use the same ten designs, ten laboratories, 800 full-client rounds and 160 development measurements per method. Uniform allocation gives every design 80 rounds; the evidence-guided loop screens all designs and fully validates six promoted candidates. Gain is evidence-guided minus uniform, in percentage points; W/T/L counts seed-level wins, ties and losses (total 4/8/0). Norman 95% CI: [4.54, 17.73]; Tahoe: [−1.00, 7.46]. The separate DTI development replay is in Appendix A.

is reported separately in Appendix A because it reuses existing trajectories rather than newly scored held-out runs.

The factorial comparison isolates participation. Holding the loop fixed, ten versus one laboratory adds +3.87 AUROC points for DTI, +16.85 AP points for proteomics, and +11.32, +8.10 and +6.97 Top-1 points for the cell settings. Appendix A reports all six combinations. In this original three-seed six-proposal comparison, direct and loop select the same checkpoints at fixed access. The allocation and proposal-history mechanisms are therefore examined with distinct controls.

5.3 HOW DO LABORATORY COUNT AND INDEPENDENT SOURCES AFFECT PERFORMANCE?

Increasing participation from one to ten laboratories raises DTI AUROC from 83.65 to 93.88 under the default recipe. With the total DTI data pool fixed, $K = 1, 2, 5, 10$ yields 85.47, 88.96, 88.05 and 93.88. The other tasks also exhibit partition sensitivity: fixed-pool PTPC AP is 34.59 at $K = 1$, 39.19 at $K = 5$ and 37.54 at $K = 10$, whereas Norman changes from 49.63 to 15.48 between $K = 1$ and $K = 10$. Repartitioning changes local optimization and, for cell tasks, the contrastive negative pool. Participation and fixed-pool curves thus characterize complementary aspects of collaborative training.

Independent collections can improve a target predictor, with gains depending on the source. DTI target-only AUROC is 85.47; Davis and Human raise it to 87.86 and 87.95, while BioSNAP and all three combined yield 82.44 and 84.30. Adding Jiang to VCC raises Top-1 from 29.49 to 33.07 (+3.58 points; paired intervention-bootstrap 95% CI [−2.11, 9.51]); combining all sources yields 27.33. In the independent-proteomics pilot, Ruprecht raises target AP from 34.98 to 35.50, and Lin plus Ruprecht gives 35.31. The Ruprecht AP change has a paired compound-bootstrap interval of [−1.46, 2.64] points. Appendix B reports every source arm and the auxiliary proteomics secondary metric. These comparisons motivate source-aware evaluation rather than indiscriminate pooling.

5.4 HOW DO THE RESEARCH MODEL AND LOOP BUDGET CHANGE THE SEARCH?

The matched six-slot comparison in Table 3 covers both research models on every endpoint. Norman/Luna gains 0.81 Top-1 points with feedback; DTI/Luna loses 0.33 AUROC points; the other eight paired endpoints tie. All twenty DTI prefix scores were independently recomputed from the same 5,505 test pairs. Thus the backend comparison measures proposal-history effects directly, separately from the fixed-slate allocation gains above.

The longer 24-slot study resolves task-dependent search dynamics. Luna reaches Norman’s common development target at slot 4 with feedback and slot 22 without it; both finish at 23.11% test Top-1. On Tahoe, direct reaches the common target earlier (5 versus 13 slots). Proteomics retains the initial design. Across the eight long-search pairs, final test scores give seven ties and one decrease with feedback (VCC/Luna, −0.22 points). Within that 36-design study, Qwen’s VCC test Top-1 falls from 31.50 at prefix 6 to 19.61 at prefix 24 as development selection improves. The full curves, acceptance decisions and per-slot configuration records in Appendix B show where feedback accelerates search and where additional proposals do not improve generalization.

Research model / method	DTI	PTPC	VCC	Norman	Tahoe
Qwen: Direct@6	94.27	37.27	31.50	22.30	25.37
Qwen: Loop@6	94.27	37.27	31.50	22.30	25.37
Luna: Direct@6	94.60	37.27	31.50	22.30	26.87
Luna: Loop@6	94.27	37.27	31.50	23.11	26.87

Table 3: Completed research-model and feedback ablation in the original 12-design, six-slot protocol. DTI uses AUROC and seed 42; PTPC uses AP and the cell endpoints use macro Top-1, all at seed 61. Each candidate receives 100 rounds on the same ten clients. The common initial design is reported in the full protocol table. Scores are multiplied by 100; bold denotes the best method within each task/backend block, including displayed ties. This table does not mix metrics across tasks or substitute the separate 36-design protocol.

6. CONCLUSION

BioCoLoop connects laboratory-local parameter fitting to an evidence-guided research loop through shared proposal, training and evaluation interfaces. The complete system leads four of five main-comparison endpoints, and compute-matched allocation improves Norman Top-1 by 10.84 points on average. Independent-source experiments demonstrate transfer gains from selected DTI and cellular collections, while laboratory-count and dual-proposer studies identify the effects of partitioning, source compatibility and research budget. The results distinguish parameter-learning gains, efficient allocation and proposal-history effects, providing an empirical basis for collaborative biological model development and more effective research feedback.

REPRODUCIBILITY STATEMENT

Section 3 specifies the research and training interfaces; Section 4 defines task inputs, data partitions, reference models and metrics. Appendix A provides the executable proposal schema, complete design library, optimization settings and matched-budget promotion rule. Appendix B specifies the laboratory-count, source-transfer and proposer-budget studies. The accompanying artifact records code and data hashes, random seeds, proposals, development trajectories, selection decisions and saved predictions. Aggregate tables are generated from completed evaluation records, with independent score recomputation for the main comparison and completed follow-up evaluations. The follow-up artifact includes all 504 attempted proposal slots and their configuration changes. These records connect each reported result to its configuration and selected checkpoint.

ETHICS STATEMENT

This computational study uses previously released biological resources under their source access conditions and collects no new measurements or participant data. Laboratory boundaries are simulated by local workers. Retaining raw data at workers avoids direct transfer to the research model; parameter updates and aggregate diagnostics would still require an explicit privacy and security policy in a multi-institutional deployment. Predictions are evaluated as research outputs, not as validated biological or therapeutic interventions.

AI USE STATEMENT

Local Qwen2.5-7B-Instruct generates proposals in the main study; the proposer sensitivity study additionally uses the service identified as gpt-5.6-luna. AI assistants contributed to hypothesis refinement, experimental protocol and software development, data processing and checks, result analysis and interpretation, literature retrieval, figures, manuscript drafting and revision, and the Chinese reading companion. The research artifacts retain the generated proposals and their measured outcomes. The authors are responsible for verifying the final methods, results, citations and manuscript, including AI-assisted material.

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APPENDIX A. UNIFORM COLLABORATIVE RESEARCH PROTOCOL

Versioned implementation. The reported study uses `core_v3.py`, `run_v3.py` and `research_v2.py`. All task families execute the same optimization, aggregation, proposal and retention source through task-specific adapters. TAPB workers use the TAPB environment; Qwen, proteomics and cell adapters use the LDA environment. Each scientific fit starts from its task-specific public or random initialization, with no reuse of earlier task-trained weights.

Shared optimization and checkpoint selection. Each round uses one local AdamW epoch, batch size 64 and gradient-norm clipping at one; local optimizer state resets between rounds. Parameters are averaged with training-example weights. If server momentum is enabled, the coordinator multiplies the previous velocity by the configured momentum coefficient and adds the difference between the averaged parameters and the previous shared parameters, then adds that velocity to the shared parameters. Floating model buffers are averaged and integer buffers use an elementwise maximum. Momentum acts on trainable parameters only. Every full-training candidate runs for 100 rounds; local development scores are collected every five rounds. Checkpoint selection maximizes the equal-client primary score, then minimizes mean development loss. The optional residual adapter has width 32 and zero-initialized output. The DTI adapter trains TAPB using frozen public protein features, the proteomic adapter trains its efficacy head from random initialization, and the cell adapters train response heads on frozen corrected scDEBART features.

Complete design library. Each identifier below compiles to one executable candidate. D00 is the fixed starting recipe; D01–D11 are available to both direct and loop search without within-arm repetition. Six proposal slots define the common budget.

Design	Learning rate	Weight decay	Momentum	Residual adapter
D00	0.001	0.001	0	no
D01	0.0001	0.001	0	no
D02	0.01	0.001	0	no
D03	0.001	0	0	no
D04	0.001	0.01	0	no
D05	0.001	0.001	0.9	no
D06	0.001	0.001	0	yes
D07	0.001	0.001	0.9	yes
D08	0.0001	0.001	0	yes
D09	0.0001	0.001	0.9	yes
D10	0.01	0.01	0	no
D11	0.001	0.01	0	yes

Proposal generation and evidence memory. The main-study proposer is Qwen2.5-7B-Instruct, using bfloat16 inference, temperature 0.7, top- p 0.9 and at most 512 new tokens. Each slot uses a deterministic seed derived from the training seed. Direct and loop comparisons share the same first proposal. Later direct proposals receive the default reference configuration and remaining untried identifiers, without evaluation scores. Loop proposals instead receive the latest retained configuration and ordered evidence cards, each containing the candidate configuration, aggregate best development metric and loss, best round, first/last training and validation records, and final status. A bounded second generation attempt repairs schema or novelty violations. Failed proposals consume a slot. Identical configurations within a seed and participation setting can reuse a completed fit while retaining their logical positions in the research traces.

Every proposal has the following executable JSON shape:

```
{
  "hypothesis": "one testable claim about an editable factor",
  "experiment": "the controlled change and incumbent comparison",
  "expected_effect": "the expected metric and diagnostic response",
  "design_id": "D00--D11"
}
```

The compiler validates `design_id` against the table above. The finalizer then attaches executed measurements and a retention decision to the proposal, creating the evidence card used in the next iteration.

Executed loop trajectories. Three ten-laboratory traces illustrate how the common finalizer operates. Values are aggregate development measurements on the 0–1 primary-metric scale, and proposal slots are one-indexed.

Study	Selected design	Primary	Loss	First direct / loop slot
PTPC, seed 42	D04	0.665833	0.521161	5 / 5
VCC, seed 43	D00	0.240350	1.607792	initial baseline
Norman, seed 42	D08	0.519000	1.538505	3 / 2

PTPC selects stronger weight decay through the BCE tie-break at equal AP. VCC retains the initial design after six alternatives. On Norman, the loop reaches the winning lower-learning-rate residual design one slot before direct search. Across all 15 ten-laboratory task–seed trajectories, the two-proposal loop is better in two, tied in thirteen and worse in none; the mean development-primary difference is +0.78 percentage points. At six proposals, direct and loop select the same checkpoints, reflecting coverage saturation in the twelve-design library.

Compute-matched evidence racing. The follow-up feedback-isolation protocol uses a ten-design fractional-factorial slate and newly executed seeds 53–55. Direct search assigns 80 rounds to every design. The loop assigns 20 screening rounds to every design, stratifies candidates by learning rate, and promotes both the strongest current candidate and the fastest-improving candidate within each stratum. Six promoted candidates restart from the common initialization for 100-round validation. Consequently, each arm executes 800 full-client rounds and 160 development measurements over the same ten hypotheses. The promotion record is finalized before any held-out response is read.

Across the resulting 12 proteomics/cell held-out comparisons, racing gives four wins, eight ties and no losses. Norman improves by 10.84 Top-1 points on average, with seed-level effects of +10.00, +11.70 and +10.81 and a hierarchical paired-bootstrap 95% interval of [4.54, 17.73]. Tahoe gives one +5.97-point seed and two ties; VCC and PTPC tie in all three seeds. Independent rescoring of every saved prediction reproduces the reported metrics exactly. A TAPB companion replay applies the same screen/promote principle to the seven-design intersection of existing traces: direct receives 525 candidate-rounds, the loop receives 520, and the development AUROC effect is +0.39 points on average (two wins, one tie).

Client and split integrity. DTI clients contain 4,932–5,058 training and 517–643 development pairs. Compound identities do not cross clients. The fixed test roles contain 5,505 random, 3,791 unseen-drug and 2,507 unseen-protein pairs. PTPC partitions 333 compound groups into 386 fitting and 105 development conditions, with 35–42 fitting conditions per client; its frozen test role contains 148 conditions. VCC, Norman and Tahoe use whole-intervention splits of 60/20/20, 70/20/27 and 180/60/67. Their ten clients own equal numbers of training and development intervention identities.

Proteomic preprocessing. For each 6h and 24h profile, the PTPC adapter applies log1p, within-profile min–max normalization over finite observations, zero filling and unit L2 normalization. The 2,048-bit Morgan vector is separately L2-normalized and scaled by 0.5. Concatenating the two 5,171-dimensional proteomic blocks with the drug vector produces a 12,390-dimensional float32 input. The ProteinTalks-derived efficacy head contains 464,193 parameters at D00. Every arm uses identical inputs, initialization policy, 100-round budget and checkpoint rule.

Full factorial results. The following tables report all six held-out arms, complete method contrasts, proposal-prefix analysis, secondary metrics, uncertainty and all three DTI test roles. Bold marks the best displayed value within each task and metric, including ties. Higher is better except BCE and cross-entropy.

Labs	Method	DTI Proteomics		Cell perturbation		
		AUROC ↑	AP ↑	VCC Top-1 ↑	Norman Top-1 ↑	Tahoe Top-1 ↑
1	Fixed model	82.83±1.22	25.05±4.56	29.94±0.00	25.56 ±0.00	19.40±0.00
1	Direct	89.89±1.02	20.13±0.07	21.09±4.38	14.15±0.00	15.42±6.21
1	Our loop	89.89±1.02	20.13±0.07	20.22±2.22	14.15±0.00	15.42±6.21
10	Fixed model	91.80±2.55	37.07 ±0.08	31.54 ±0.08	15.48±0.00	25.87 ±1.72
10	Direct	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
10	Our loop	93.76 ±1.11	36.99±0.22	31.54 ±0.08	22.25±0.45	22.39±2.59
Completed seeds ($n/3$)		3/3	3/3	3/3	3/3	3/3

Table A.1: Six-arm ablation of the task-reference models. Entries are held-out mean $\pm$ sample SD across completed training/search seeds, multiplied by 100. The data partitions remain fixed; this SD measures run-to-run variation, not dataset uncertainty. DTI uses the random held-out endpoint. Bold marks all displayed maxima.

Held-out contrast	DTI	Proteomics	VCC	Norman	Tahoe
Loop – fixed (1 lab)	+7.06	-4.91	-9.71	-11.41	-3.98
Loop – fixed (10 labs)	+1.96	-0.08	+0.00	+6.77	-3.48
Loop – direct (1 lab)	+0.00	+0.00	-0.87	+0.00	+0.00
Loop – direct (10 labs)	+0.00	+0.00	+0.00	+0.00	+0.00
10 – 1 labs (both fixed)	+8.97	+12.02	+1.61	-10.07	+6.47
10 – 1 labs (both direct)	+3.87	+16.85	+10.45	+8.10	+6.97
10 – 1 labs (both loop)	+3.87	+16.85	+11.32	+8.10	+6.97
Completed seeds ($n/3$)	3/3	3/3	3/3	3/3	3/3

Table A.2: Complete decomposition into design-search improvement (loop minus fixed), feedback-specific improvement (loop minus matched-budget direct), and laboratory-participation improvement (ten minus one lab with loop fixed). Mean paired held-out differences are in percentage points over the three seeds and are computed before rounding.

Proposal budget	Loop better	Tie	Direct better	Mean Δ primary points
	task-seed trajectories			
1	0	15	0	+0.00
2	2	13	0	+0.78
3	0	14	1	-0.04
4	0	15	0	+0.00
5	0	14	1	-0.02
6	0	15	0	+0.00

Table A.3: Executed-prefix analysis of all 15 ten-laboratory development trajectories at common proposal budgets. At two proposals, feedback-guided search improves TAPB seed 44 by 4.84 points and Norman seed 42 by 6.90 points, with thirteen ties and no regressions.

Table A.4: Held-out secondary metrics for the task-reference models, averaged over seeds 42–44. Bold indicates all tied maxima, or minima for loss.

Method	Labs	Metric 1	Metric 2	Metric 3	Metric 4
DTI ($n = 3$)		AUROC	AP	MCC	BCE
Fixed model	1	0.8283	0.8157	0.4995	0.5237
Direct	1	0.8989	0.8777	0.6571	0.6102
Our loop	1	0.8989	0.8777	0.6571	0.6102
Fixed model	10	0.9180	0.8936	0.7239	0.3654
Direct	10	0.9376	0.9273	0.7371	0.3659
Our loop	10	0.9376	0.9273	0.7371	0.3659
Proteomics ($n = 3$)		AP	AUROC	BCE	Acc.
Fixed model	1	0.2505	0.6549	0.4265	0.8514
Direct	1	0.2013	0.6180	0.4434	0.8176
Our loop	1	0.2013	0.6180	0.4434	0.8176
Fixed model	10	0.3707	0.7406	0.3978	0.8514
Direct	10	0.3699	0.7387	0.3983	0.8514
Our loop	10	0.3699	0.7387	0.3983	0.8514
VCC ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.3361	0.5653	0.7389	1.5451
Direct	1	0.2148	0.4710	0.6278	1.6082
Our loop	1	0.2074	0.4694	0.6426	1.6086
Fixed model	10	0.3398	0.5634	0.6944	1.5814
Direct	10	0.3398	0.5634	0.6944	1.5814
Our loop	10	0.3398	0.5634	0.6944	1.5814
Norman ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.2556	0.4850	0.6393	1.5874
Direct	1	0.1415	0.4212	0.5835	1.6039
Our loop	1	0.1415	0.4212	0.5835	1.6039
Fixed model	10	0.1548	0.4526	0.6563	1.5806
Direct	10	0.2225	0.5007	0.7096	1.5883
Our loop	10	0.2225	0.5007	0.7096	1.5883
Tahoe ($n = 3$)		Micro Top-1	MRR	Top-3	CE
Fixed model	1	0.1940	0.4500	0.5821	1.6308
Direct	1	0.1542	0.4260	0.5672	1.6190
Our loop	1	0.1542	0.4260	0.5672	1.6190
Fixed model	10	0.2587	0.5020	0.6517	1.6061
Direct	10	0.2239	0.4799	0.6169	1.6075
Our loop	10	0.2239	0.4795	0.6119	1.6075

Table A.5: Paired cluster-bootstrap 95% intervals for each seed and main contrast, in percentage points. Intervals condition on the selected trained models; DTI random-endpoint rows resample drug clusters.

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – fixed (10 labs)	42	+0.33	[-0.06, +0.69]
DTI	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	42	+12.51	[+11.35, +13.71]
DTI	10 – 1 labs (both direct)	42	+5.78	[+5.02, +6.57]
DTI	10 – 1 labs (both loop)	42	+5.78	[+5.02, +6.57]
DTI	Loop – fixed (10 labs)	43	+0.54	[-0.08, +1.11]
DTI	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	43	+7.80	[+6.82, +8.82]
DTI	10 – 1 labs (both direct)	43	+1.65	[+0.90, +2.38]
DTI	10 – 1 labs (both loop)	43	+1.65	[+0.90, +2.38]
DTI	Loop – fixed (10 labs)	44	+5.01	[+4.25, +5.71]
DTI	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	44	+6.60	[+5.61, +7.74]
DTI	10 – 1 labs (both direct)	44	+4.17	[+3.45, +4.93]
DTI	10 – 1 labs (both loop)	44	+4.17	[+3.45, +4.93]
Proteomics	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	42	+13.16	[-0.39, +26.02]
Proteomics	10 – 1 labs (both direct)	42	+16.91	[+3.37, +31.02]
Proteomics	10 – 1 labs (both loop)	42	+16.91	[+3.37, +31.02]
Proteomics	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	43	+15.98	[+1.29, +29.00]
Proteomics	10 – 1 labs (both direct)	43	+17.03	[+3.34, +31.44]
Proteomics	10 – 1 labs (both loop)	43	+17.03	[+3.34, +31.44]
Proteomics	Loop – fixed (10 labs)	44	-0.24	[-3.40, +2.33]
Proteomics	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	44	+6.93	[-3.43, +15.60]
Proteomics	10 – 1 labs (both direct)	44	+16.63	[+3.02, +31.08]
Proteomics	10 – 1 labs (both loop)	44	+16.63	[+3.02, +31.08]
VCC	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	42	+1.70	[-5.36, +8.58]
VCC	10 – 1 labs (both direct)	42	+9.90	[-2.88, +23.04]
VCC	10 – 1 labs (both loop)	42	+9.90	[-2.88, +23.04]
VCC	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	43	+4.83	[-5.50, +16.49]
VCC	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	43	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	43	+15.08	[+2.94, +27.78]
VCC	10 – 1 labs (both loop)	43	+10.24	[-2.74, +23.49]
VCC	Loop – fixed (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	44	-7.44	[-17.23, +1.33]
VCC	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	44	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	44	+6.38	[-4.38, +19.13]
VCC	10 – 1 labs (both loop)	44	+13.82	[+2.54, +26.34]
Norman	Loop – fixed (10 labs)	42	+7.26	[-9.41, +25.41]
Norman	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	42	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	42	+8.59	[-8.30, +26.00]
Norman	10 – 1 labs (both loop)	42	+8.59	[-8.30, +26.00]
Norman	Loop – fixed (10 labs)	43	+6.37	[-11.12, +24.81]

Task	Contrast	Seed	Difference	95% interval
Norman	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	43	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	43	+7.70	[-9.33, +25.85]
Norman	10 – 1 labs (both loop)	43	+7.70	[-9.33, +25.85]
Norman	Loop – fixed (10 labs)	44	+6.67	[-10.37, +24.81]
Norman	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	44	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	44	+8.00	[-8.52, +25.93]
Norman	10 – 1 labs (both loop)	44	+8.00	[-8.52, +25.93]
Tahoe	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	42	+4.48	[-5.97, +14.93]
Tahoe	10 – 1 labs (both direct)	42	+10.45	[-1.49, +22.39]
Tahoe	10 – 1 labs (both loop)	42	+10.45	[-1.49, +22.39]
Tahoe	Loop – fixed (10 labs)	43	-2.99	[-7.46, +0.00]
Tahoe	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	43	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	43	+13.43	[+1.49, +23.88]
Tahoe	10 – 1 labs (both loop)	43	+13.43	[+1.49, +23.88]
Tahoe	Loop – fixed (10 labs)	44	-7.46	[-19.40, +4.48]
Tahoe	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	44	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	44	-2.99	[-17.91, +11.94]
Tahoe	10 – 1 labs (both loop)	44	-2.99	[-17.91, +11.94]

Table A.6: All three frozen TAPB held-out endpoints under the committed common protocol. Scores are completed-seed means multiplied by 100; N/A means no complete independently verified six-arm evaluation. The main DTI endpoint is random held-out.

Endpoint	Method	Labs	AUROC	AP
Random	Fixed model	1	82.83	81.57
Random	Direct	1	89.89	87.77
Random	Our loop	1	89.89	87.77
Random	Fixed model	10	91.80	89.36
Random	Direct	10	93.76	92.73
Random	Our loop	10	93.76	92.73
Unseen drug	Fixed model	1	81.96	85.78
Unseen drug	Direct	1	88.39	89.35
Unseen drug	Our loop	1	88.39	89.35
Unseen drug	Fixed model	10	90.06	90.40
Unseen drug	Direct	10	91.65	92.58
Unseen drug	Our loop	10	91.65	92.58
Unseen protein	Fixed model	1	76.40	78.75
Unseen protein	Direct	1	83.26	80.53
Unseen protein	Our loop	1	83.26	80.53
Unseen protein	Fixed model	10	85.80	78.92

Endpoint	Method	Labs	AUROC	AP
Unseen protein	Direct	10	87.63	85.07
Unseen protein	Our loop	10	87.63	85.07

Table A.7: Post-hoc TAPB paired AUROC/AP differences and 95% intervals from 1,000 fixed-seed cluster-bootstrap draws. Random and unseen-drug endpoints use drug clusters; unseen-protein uses protein clusters. All three endpoints, six arms and prespecified contrasts are shown.

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Random / 42	Loop — fixed (10 labs)	+0.33	[-0.06, +0.69]	+1.40	[+0.76, +2.05]
Random / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	10 — 1 labs (fixed)	+12.51	[+11.35, +13.71]	+10.56	[+9.10, +12.11]
Random / 42	10 — 1 labs (direct)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	10 — 1 labs (loop)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	Our harness — fixed (1 lab)	+12.84	[+11.82, +14.01]	+11.96	[+10.56, +13.45]
Random / 42	Our harness — direct (1 lab)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Unseen drug / 42	Loop — fixed (10 labs)	+0.10	[-0.47, +0.67]	+1.06	[+0.42, +1.66]
Unseen drug / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	10 — 1 labs (fixed)	+12.30	[+10.77, +13.98]	+7.17	[+5.82, +8.72]
Unseen drug / 42	10 — 1 labs (direct)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	10 — 1 labs (loop)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	Our harness — fixed (1 lab)	+12.40	[+10.94, +13.85]	+8.22	[+6.97, +9.66]
Unseen drug / 42	Our harness — direct (1 lab)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen protein / 42	Loop — fixed (10 labs)	+0.03	[-5.53, +4.28]	+4.12	[-4.10, +8.22]
Unseen protein / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	10 — 1 labs (fixed)	+20.02	[+0.16, +63.83]	+8.14	[-8.27, +51.77]
Unseen protein / 42	10 — 1 labs (direct)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	10 — 1 labs (loop)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	Our harness — fixed (1 lab)	+20.05	[+1.84, +61.27]	+12.25	[-3.01, +53.19]
Unseen protein / 42	Our harness — direct (1 lab)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Random / 43	Loop — fixed (10 labs)	+0.54	[-0.08, +1.11]	+0.44	[-0.54, +1.44]
Random / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	10 — 1 labs (fixed)	+7.80	[+6.82, +8.82]	+7.68	[+6.32, +9.13]
Random / 43	10 — 1 labs (direct)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	10 — 1 labs (loop)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	Our harness — fixed (1 lab)	+8.35	[+7.29, +9.37]	+8.12	[+6.53, +9.76]
Random / 43	Our harness — direct (1 lab)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Unseen drug / 43	Loop — fixed (10 labs)	-0.51	[-1.44, +0.29]	-1.10	[-2.16, -0.18]
Unseen drug / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	10 — 1 labs (fixed)	+6.68	[+5.33, +8.25]	+4.66	[+3.33, +6.36]
Unseen drug / 43	10 — 1 labs (direct)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	10 — 1 labs (loop)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	Our harness — fixed (1 lab)	+6.17	[+4.85, +7.48]	+3.57	[+2.10, +5.09]
Unseen drug / 43	Our harness — direct (1 lab)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen protein / 43	Loop — fixed (10 labs)	-0.96	[-7.24, +3.07]	-1.83	[-21.55, +7.54]
Unseen protein / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	10 — 1 labs (fixed)	+4.79	[-6.61, +28.12]	+0.41	[-16.48, +31.69]
Unseen protein / 43	10 — 1 labs (direct)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	10 — 1 labs (loop)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	Our harness — fixed (1 lab)	+3.83	[-5.83, +22.32]	-1.42	[-14.49, +16.88]
Unseen protein / 43	Our harness — direct (1 lab)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Random / 44	Loop — fixed (10 labs)	+5.01	[+4.25, +5.71]	+8.28	[+7.07, +9.56]
Random / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	10 — 1 labs (fixed)	+6.60	[+5.61, +7.74]	+5.12	[+3.71, +6.80]
Random / 44	10 — 1 labs (direct)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	10 — 1 labs (loop)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	Our harness — fixed (1 lab)	+11.60	[+10.60, +12.65]	+13.40	[+11.88, +15.08]
Random / 44	Our harness — direct (1 lab)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Unseen drug / 44	Loop — fixed (10 labs)	+5.18	[+4.09, +6.19]	+6.57	[+5.28, +7.85]
Unseen drug / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	10 — 1 labs (fixed)	+5.32	[+3.81, +6.97]	+2.04	[+0.38, +3.90]
Unseen drug / 44	10 — 1 labs (direct)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	10 — 1 labs (loop)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Unseen drug / 44	Our harness — fixed (1 lab)	+10.50	[+9.20, +11.87]	+8.61	[+7.10, +10.29]
Unseen drug / 44	Our harness — direct (1 lab)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen protein / 44	Loop — fixed (10 labs)	+6.42	[-2.79, +17.37]	+16.15	[-5.79, +26.26]
Unseen protein / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	10 — 1 labs (fixed)	+3.39	[-16.02, +33.81]	-8.03	[-26.82, +16.99]
Unseen protein / 44	10 — 1 labs (direct)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	10 — 1 labs (loop)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	Our harness — fixed (1 lab)	+9.81	[-2.07, +34.89]	+8.12	[-12.06, +22.71]
Unseen protein / 44	Our harness — direct (1 lab)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]

Uncertainty analysis. The DTI analysis is computed after training and held-out scoring, using paired cluster bootstrap with 1,000 fixed-seed resamples of retained predictions. It groups by released drug identity on random/unseen-drug endpoints and by protein sequence on unseen-protein. Other task intervals use their natural compound or intervention grouping. Resamples are shared within each method contrast. These intervals characterize sample-level variation within each selected fit; between-training-seed variation is reported through the three-seed means and sample SD.

Verification. CPU unit and integration tests cover the common trainer, task adapters, proposal generation, frozen evaluation and cell attention-mask correction. Publication checks bind table cells to completed arms, model versions, checkpoint seals and independent score recomputation. The snapshot includes receipt hashes, aggregate metrics and the complete proposal histories used in this manuscript.

APPENDIX B. LABORATORY, SOURCE AND PROPOSER SENSITIVITY

Completed study design. This appendix reports completed follow-up evaluations with the same target scorers as the main study. Laboratory-count experiments cover all five endpoints at $K \in \{1, 2, 5, 10\}$ under both increasing participation and a fixed total data pool. DTI uses seed 61; the lighter proteomics and cell experiments use seeds 61–63. Independent-source DTI and proteomics use seed 61, and external-source VCC uses seeds 61–63. The two-proposer study has two separate protocols: a 12-design, six-slot study on all five endpoints (DTI seed 42; other endpoints seed 61), and a 36-design, 24-slot study on the four lighter endpoints at seed 61. Each candidate receives 100 training rounds. The latter changes the design space as well as the search budget, so budget comparisons are made within a protocol. Main-table results retain their original seeds 42–44 and protocol.

B.1 NUMBER OF LABORATORIES

The participation experiment uses a fixed nested set of the first K original clients, increasing both training and development access. The fixed-pool experiment merges the complete ten-client pool into K training groups while retaining equal-weight evaluation over the original ten development panels. Thus its selection population stays fixed as training partitions change. The identical $K = 10$ fit is reused by both comparisons. Every unique fit uses the default design and 100 rounds. The cell objective forms contrastive negatives within each local training group, so repartitioning changes the negative pool as well as optimization steps. These comparisons characterize the implemented training pipeline at different collaboration scales. Zero between-seed SD can occur when a fixed pretrained head receives an identical single-batch update; it describes training variability on these fixed partitions.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	83.65	84.43	89.62	93.88
PTPC	AP	26.03 ± 5.57	37.38 ± 0.60	29.63 ± 2.50	37.54 ± 0.27
VCC	Macro Top-1	30.02 ± 0.78	39.56 ± 0.11	31.88 ± 0.00	31.48 ± 0.16
Norman	Macro Top-1	25.56 ± 0.00	18.22 ± 0.00	19.68 ± 1.11	15.48 ± 0.00
Tahoe	Macro Top-1	19.40 ± 0.00	25.37 ± 0.00	20.90 ± 0.00	26.37 ± 0.86

Table B.1: Increasing participation. A nested set of the original ten client partitions participates, so increasing K also adds training data. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

Task	Metric	$K = 1$	$K = 2$	$K = 5$	$K = 10$
DTI	AUROC	85.47	88.96	88.05	93.88
PTPC	AP	34.59 ± 5.66	37.36 ± 1.89	39.19 ± 0.43	37.54 ± 0.27
VCC	Macro Top-1	29.49 ± 0.00	28.80 ± 0.20	31.06 ± 0.22	31.48 ± 0.16
Norman	Macro Top-1	49.63 ± 4.20	19.56 ± 0.20	26.12 ± 0.11	15.48 ± 0.00
Tahoe	Macro Top-1	22.39 ± 0.00	21.39 ± 0.86	20.40 ± 0.86	26.37 ± 0.86

Table B.2: Fixed data pool. All target data are retained and merged into K clients; only the partition count changes. The default design and 100-round trainer are fixed. DTI uses seed 61; the four light endpoints report mean $\pm$ sample SD over seeds 61–63. Scores are metric $\times 100$; bold denotes the best K within each row, including displayed ties.

B.2 INDEPENDENT SOURCES AND TARGET EVALUATION

For DTI and cell transfer, the entire target training pool is one study-level client. Each auxiliary collection contributes a separate local update through sample-weighted aggregation. Only the original target development panels select the checkpoint; every configuration is evaluated on the same target test set. Target-only, individual-source and combined-source configurations start from the same task initialization.

DTI auxiliaries are BioSNAP, Davis and Human from the pinned TAPB release, with 19,092, 1,721 and 2,929 retained training pairs after drug–protein-pair de-duplication against target splits and preceding sources. All five DTI source configurations complete the same 100-round schedule. VCC auxiliary data comprise Replogle K562, Nadig HepG2 and Jiang IFNB/K562, with respectively 60, 60 and 58 retained training interventions (Replogle et al., 2022; Nadig et al., 2025; Jiang et al., 2025a). Source feature panels preserve measured genes, and target development/test intervention identities are excluded from source training. The source manifest records retained subsets, feature intersections and exclusions.

Target	Additional source	Metric	Score $\uparrow$	Δ (pp)
DTI	Target only	AUROC	85.47	+0.00
DTI	+ BioSNAP	AUROC	82.44	-3.03
DTI	+ Davis	AUROC	87.86	+2.39
DTI	+ Human	AUROC	87.95	+2.48
DTI	+ All three	AUROC	84.30	-1.18
VCC	Target only	Macro Top-1	29.49 $\pm$ 0.00	+0.00 $\pm$ 0.00
VCC	+ Replogle	Macro Top-1	27.12 $\pm$ 0.08	-2.37 $\pm$ 0.08
VCC	+ Nadig	Macro Top-1	26.94 $\pm$ 0.18	-2.55 $\pm$ 0.18
VCC	+ Jiang	Macro Top-1	33.07 $\pm$ 0.00	+3.58 $\pm$ 0.00
VCC	+ All three	Macro Top-1	27.33 $\pm$ 0.00	-2.15 $\pm$ 0.00
PTPC auxiliary	Target only	AP	34.98	+0.00
PTPC auxiliary	+ Lin	AP	35.01	+0.03
PTPC auxiliary	+ Ruprecht	AP	35.50	+0.52
PTPC auxiliary	+ Both	AP	35.31	+0.34

Table B.3: Independent-source transfer on the same target heldout set, including every prespecified source and their combination. DTI (seed 61) adds BioSNAP, Davis or Human; VCC (seeds 61–63, mean $\pm$ SD) adds Replogle, Nadig or Jiang. The PTPC auxiliary multi-task model (seed 61) adds Lin or Ruprecht and is compared only with its matched auxiliary target-only model. PTPC has two independent external studies, not three. Changes are paired against target-only training; bold marks the best score within a target/model block. Scores are multiplied by 100.

The cell study reports mean and SD across three seeds together with a paired intervention-bootstrap analysis. Jiang improves target Top-1 by 3.58 points; its 95% bootstrap interval is $[-2.11, 9.51]$. This positive point estimate motivates further source-aware evaluation, while the complete table also exposes sources whose updates reduce the target metric.

Independent proteomics pilot. Two independent lung-cancer studies provide auxiliary measurements: Lin et al. (Lin et al., 2026) and Ruprecht et al. (Ruprecht et al., 2020). Their continuous response labels differ from the binary target endpoint. A shared masked-protein encoder of width 64 and drug encoder of width 32 therefore feed source-specific prediction heads. Target fitting uses its binary efficacy objective; Lin predicts continuous CellTiter-Glo response, and Ruprecht predicts an interval-valued potency outcome. Target data retain the original ten clients; Lin and Ruprecht each add one study-level client. Shared encoders use sample-weighted aggregation, whereas each private head is updated only by its own source clients. The common local-worker and coordinator interfaces are reused. This adapter is compared against its own target-only control, separately from the main-table PTPC predictor.

The Lin subset contains 28 drug conditions (20 source-training, eight source-development), spanning five training and two development drugs. Inputs are 24-hour, 10- μ M protein log-fold changes, and the response is the mean of the three released CellTiter-Glo ratio measurements. The HCT16/HCT116 alias is excluded from auxiliary training. The Ruprecht subset contains 109 matched drug conditions (85 training, 24 development; 17 and five drugs respectively). Protein log-fold changes at 24 hours are paired with CellTiter-Glo EC50 values in $\log_{10}$ nM: 49 exact, 18 approximate and 42 right-censored outcomes. Squared distance to the encoded potency interval is the loss. The adapter treats explicit lower bounds and values at or above 10,000 nM as right-censored and encodes approximate values with a factor-of-two interval; these are analysis conventions. Profiling dose is not a model input; source acquisition nevertheless uses dose conditions chosen in relation to drug response. The

target retains 386 training and 105 development examples, with 148 held-out observations across 92 compounds and 22 positives.

All four source settings receive the same 100-round procedure. A two-proposal Luna loop is also executed independently for each setting. These 12 fits and eight valid proposal calls produce eight fixed/loop evaluations. Fixed and loop selection have identical held-out ranking scores in each source arm (Table B.4). The source effect and the proposal-history effect are thus measured separately.

Training sources	Selection	AP $\uparrow$	Δ vs target only (pp)
Target only	Fixed	34.98	+0.00
Target only	Loop	34.98	+0.00
+ Lin	Fixed	35.01	+0.03
+ Lin	Loop	35.01	+0.03
+ Ruprecht	Fixed	35.50	+0.52
+ Ruprecht	Loop	35.50	+0.52
+ Both	Fixed	35.31	+0.34
+ Both	Loop	35.31	+0.34

Table B.4: Independent-source proteomics auxiliary pilot, seed 61. Fixed design and a two-proposal Luna loop use the same shared-encoder/source-specific-head model and 100-round training allocation. Every source arm and both selection modes are retained. The heldout AP gain from this short loop is zero in all four source conditions; source-transfer gains are a different comparison.

Target-only AP/AUROC are 34.98/74.75. Adding Lin yields 35.01/74.78, Ruprecht 35.50/74.28 and both 35.31/74.17. Paired compound bootstrap (1,000 resamples) gives 95% intervals for the AP changes of $[-0.20, 0.28]$, $[-1.46, 2.64]$ and $[-1.62, 2.49]$ points, respectively. These results measure transfer from two independent external studies. The three cell-line contexts below answer a separate, within-study question.

Within-study proteomic contexts. HCC1143, HCC1806 and MDA-MB-453-1 are drawn from the mtPTDS supplementary collection (Sun et al., 2026), each with 47 retained drug conditions and 3,501 proteins shared with the target panel. They are three biological contexts within one study. Target development/test compounds are excluded from source training. These experiments use the original proteomic adapter and seeds 61–63.

Additional context	AP $\uparrow$	Δ AP (pp)	Paired 95% CI
Target only	34.59 $\pm$ 5.66	+0.00 $\pm$ 0.00	[0.00, 0.00]
+ HCC1143	34.94 $\pm$ 5.15	+0.35 $\pm$ 0.56	[-1.22, 1.60]
+ HCC1806	35.23 $\pm$ 5.35	+0.64 $\pm$ 0.32	[-0.78, 1.77]
+ MDA-MB-453-1	35.20 $\pm$ 5.31	+0.61 $\pm$ 0.44	[-0.85, 2.04]
+ All three	35.47 $\pm$ 6.36	+0.87 $\pm$ 8.89	[-2.44, 3.99]

Table B.5: Within-study PTPC context transfer, reported separately from independent-source transfer. HCC1143, HCC1806 and MDA-MB-453-1 belong to the same mtPTDS study. Values are mean $\pm$ sample SD over seeds 61–63, multiplied by 100. Confidence intervals resample target compounds while retaining paired seed predictions. These contexts are not additional independent studies; the original model and results are unchanged.

B.3 RESEARCH-MODEL AND SIX-SLOT PROPOSAL COMPARISON

The short study uses the 12 executable designs of Appendix A. Direct search receives untried identifiers without measured scores; the loop additionally receives accumulated development evidence. Both modes share initialization and the first proposal. Each slot allows two attempts to return the four-field proposal schema; failed slots consume budget and remain in the trace. Every valid candidate restarts from the common task/seed initialization and is fitted for 100 rounds. Prefixes at 0, 1, 2, 4 and 6 slots are evaluated after the trajectory is complete. An exact fit cache can reuse an identical configuration while retaining its logical training-budget charge.

Qwen uses the local sampling settings in Appendix A. The second proposer is the gpt-5.6-luna service, accessed in a text-only session with tool use disabled; request metadata and responses are retained. The comparison holds task messages, editable designs, training procedure and proposal budget fixed across backends. Service-level decoding controls and parameter count are not exposed by the Luna interface.

Task	Proposer	Fixed	Direct@6	Loop@6
DTI	Luna	94.27	94.60	94.27
DTI	Qwen	94.27	94.27	94.27
PTPC	Luna	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27
VCC	Luna	31.50	31.50	31.50
VCC	Qwen	31.50	31.50	31.50
Norman	Luna	15.48	22.30	23.11
Norman	Qwen	15.48	22.30	22.30
Tahoe	Luna	25.37	26.87	26.87
Tahoe	Qwen	25.37	25.37	25.37

Table B.6: Original 12-design/six-slot heldout results. DTI seed 42; four light endpoints seed 61. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
DTI	Luna	94.71	0/0	6/0	6/0	6/6
DTI	Qwen	94.71	0/0	0/0	0/0	6/6
PTPC	Luna	66.72	0/0	0/0	0/0	6/6
PTPC	Qwen	66.72	0/0	0/0	0/0	6/5
VCC	Luna	24.04	3/6	3/6	3/6	6/6
VCC	Qwen	23.65	0/0	0/0	0/0	6/6
Norman	Luna	51.70	5/5	5/5	5/5	6/6
Norman	Qwen	51.70	3/4	3/4	3/4	6/6
Tahoe	Luna	30.00	0/0	0/0	6/3	6/6
Tahoe	Qwen	30.00	0/0	0/0	0/0	4/4

Table B.7: Original 12-design/six-slot search dynamics. DTI seed 42; four light endpoints seed 61. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

All five endpoints have both backends and both search modes. Among the ten direct/loop pairs, Norman/Luna improves by 0.81 Top-1 points, DTI/Luna decreases by 0.33 AUROC points and eight pairs tie. The twenty DTI prefix scores were independently recomputed from saved predictions for all 5,505 test pairs, with checkpoint and output hashes verified. The curves below show the complete prespecified prefix sequence for every endpoint, alongside its development selection trajectory.

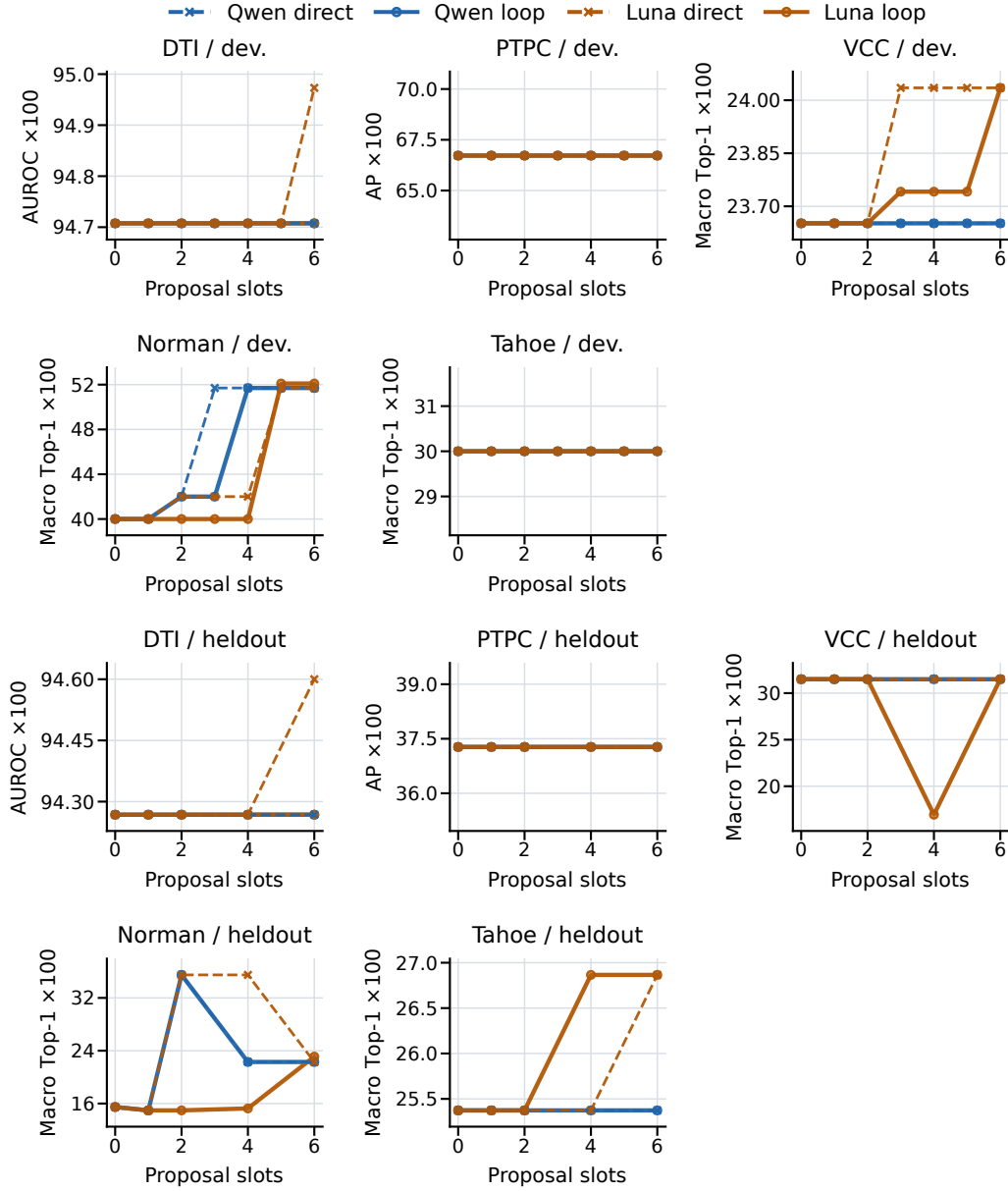


Figure 3: Development selection (upper panels) and held-out performance (lower panels) in the 12-design, six-slot protocol across all five endpoints. Each valid proposal receives the same full fitting schedule; failed slots still consume research budget. Slot 0 is initialization. Held-out curves evaluate the development-selected checkpoints at every prespecified prefix.

B.4 EXTENDED SEARCH AND TASK-DEPENDENT SATURATION

The extended study increases the design library to 36 configurations: three learning rates, three weight decays, two server-momentum values and two residual-head settings. It runs 24 proposal slots for both backends and modes on PTPC, VCC, Norman and Tahoe. Prefixes are 0, 1, 2, 4, 6, 8, 12, 16, 20 and 24. The settings for initialization, full candidate fitting, development selection and invalid-slot accounting are otherwise unchanged. DTI’s backend comparison is the completed six-slot study in B.3; it is not part of this extended protocol.

Task	Proposer	Fixed	Direct@6	Loop@6	Direct@24	Loop@24
PTPC	Luna	37.27	37.27	37.27	37.27	37.27
PTPC	Qwen	37.27	37.27	37.27	37.27	37.27
VCC	Luna	31.50	16.94	16.94	20.23	20.01
VCC	Qwen	31.50	31.50	31.50	19.61	19.61
Norman	Luna	15.48	35.48	23.11	23.11	23.11
Norman	Qwen	15.48	22.30	22.30	22.30	22.30
Tahoe	Luna	25.37	13.43	26.87	13.43	13.43
Tahoe	Qwen	25.37	26.87	26.87	13.43	13.43

Table B.8: Extended 36-design/24-slot heldout results. Four light endpoints, seed 61; no native DTI 24-slot result is implied. All displayed budgets are prefixes of this single protocol; they are not selected by test performance. Every candidate uses 100 rounds. Scores use each original task metric $\times 100$; bold marks the best displayed score within a row, including ties.

Task	Proposer	Dev. target	First D/L	Last gain D/L	Last retain D/L	Valid D/L
PTPC	Luna	66.72	0/0	0/0	0/0	24/24
PTPC	Qwen	66.72	0/0	0/0	0/0	19/14
VCC	Luna	27.68	16/9	16/10	16/10	24/24
VCC	Qwen	24.83	18/23	18/23	18/23	11/16
Norman	Luna	52.10	22/4	22/4	22/7	24/24
Norman	Qwen	51.70	2/5	2/5	22/19	15/12
Tahoe	Luna	35.00	5/13	5/13	5/13	24/24
Tahoe	Qwen	35.00	18/14	18/14	18/14	14/15

Table B.9: Extended 36-design/24-slot search dynamics. Four light endpoints, seed 61; no native DTI 24-slot result is implied. D/L means direct/loop. The common target is the lower of the two terminal development primary scores, so it is attainable by both runs; first records its earliest attainment. Last gain is the last strictly improving primary-score slot; last retain also includes loss-tiebreak selections. These are retrospective diagnostics, not stopping rules. Slot 0 is initialization.

Luna produces valid designs in all 192 attempted slots, and Qwen in 116 of 192. Across 16 mode trajectories, the primary development score is unchanged in the final eight slots in 12 cases; the full score/loss selection key is unchanged in ten. Each trajectory retains 11–24 unvisited designs. A plateau therefore characterizes finite-budget search with a particular backend, rather than exhaustion of the design space.

Table B.9 measures the first slot attaining the smaller terminal development score within each direct/loop pair, a common target reached by both modes. Luna reaches Norman’s common target earlier with feedback (4 versus 22 slots), whereas direct search reaches Tahoe’s target earlier (5 versus 13). Proteomics retains the initial design. Table B.10 connects Norman’s trajectory to concrete configuration changes: reducing the learning rate to 10^{-4} gives an early development gain, and removing the residual head reaches the final primary-score target. Subsequent loss-tiebreak selections can change the retained design without increasing that primary score.

The eight terminal held-out pairs give seven ties and one decrease with feedback (VCC/Luna, -0.22 points). Within this extended protocol, Qwen’s VCC held-out Top-1 decreases from 31.50 at slot 6 to 19.61 at slot 24 while development selection improves. This separates the speed of finding a strong development design from generalization of the selected predictor. The complete development and held-out curves support task-specific budget analysis alongside the proposal-validity and remaining-space diagnostics.

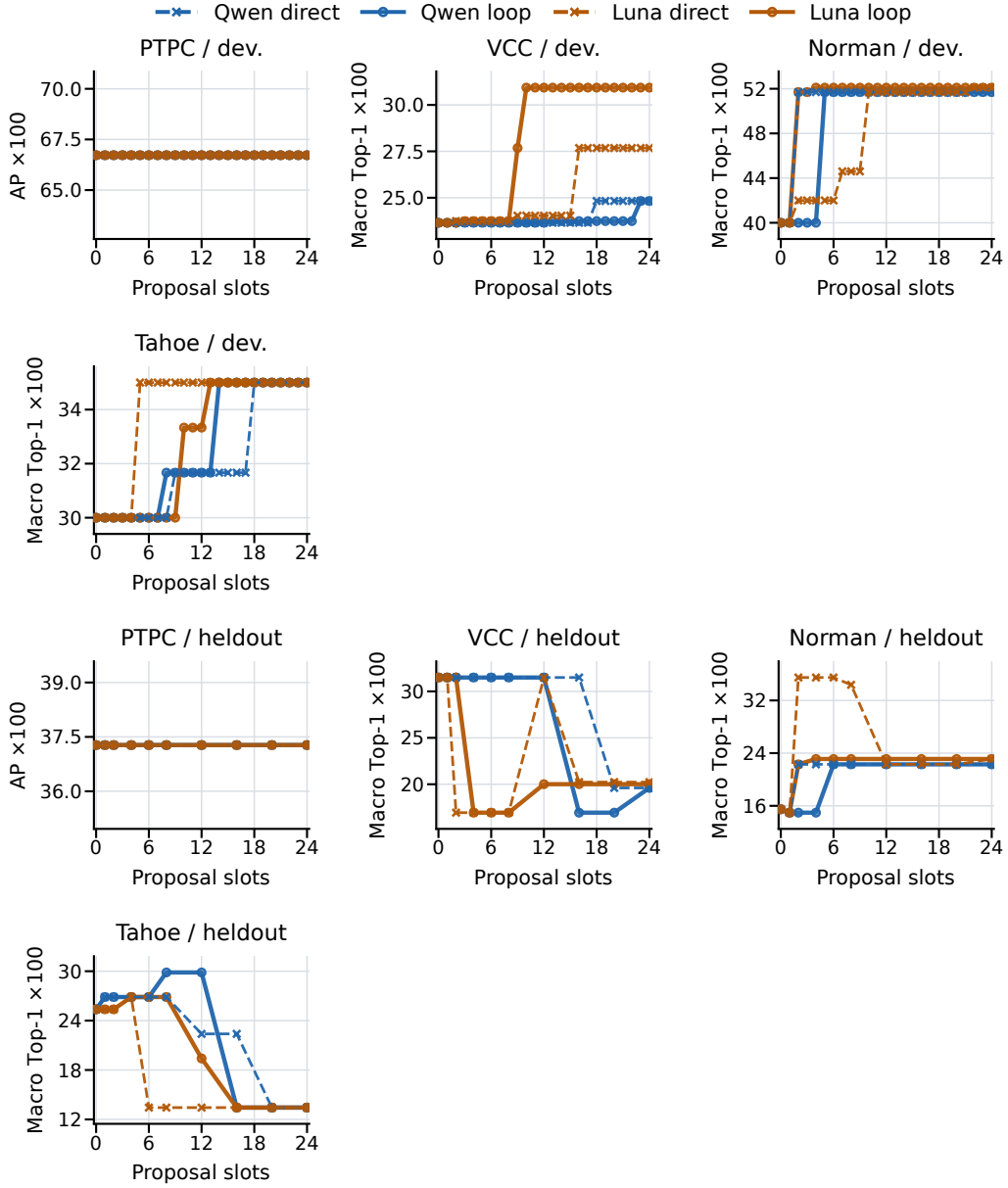


Figure 4: Development selection (upper panels) and held-out performance (lower panels) in the separate 36-design, 24-slot protocol on four endpoints. Curves are paired within the same backend, initialization and design space. Score plateaus may still include loss-tiebreak selections. All prespecified prefixes are displayed. Six-slot points here belong to this 36-design protocol, not the separate 12-design experiment.

B.5 EXECUTED DESIGNS AND PROPOSAL TRACES

Tables B.11 and B.12 decode the two design libraries. Tables B.13 and B.14 retain all 504 attempted proposal slots and their acceptance decisions: 120 slots in the short protocol and 384 in the extended protocol. The accompanying machine-readable trace additionally records each proposal’s full configuration, changes from the incumbent and measured development score. Together with the prefix curves, these records connect research decisions to both the executable intervention and its observed outcome.

Mode	Slot	Design	Retained configuration	Dev. Top-1
Direct	0	Initial	Common initial design	40.00
Direct	1	L1W1M0R1	lr=0.001, wd=0.001, m=0, R=1	40.00
Direct	2	L1W1M1R1	lr=0.001, wd=0.001, m=0.9, R=1	42.00
Direct	7	L1W2M1R1	lr=0.001, wd=0.01, m=0.9, R=1	44.60
Direct	10	L0W0M0R1	lr=0.0001, wd=0, m=0, R=1	51.70
Direct	11	L0W1M0R1	lr=0.0001, wd=0.001, m=0, R=1	51.70
Direct	22	L0W1M0R0	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	0	Initial	Common initial design	40.00
Loop	1	L1W1M0R1	lr=0.001, wd=0.001, m=0, R=1	40.00
Loop	2	L0W1M0R1	lr=0.0001, wd=0.001, m=0, R=1	51.70
Loop	4	L0W1M0R0	lr=0.0001, wd=0.001, m=0, R=0	52.10
Loop	7	L0W0M0R0	lr=0.0001, wd=0, m=0, R=0	52.10

Table B.10: All retained proposals for the Norman/Luna extended 36-design, 24-slot study, seed 61. The two modes share initialization and first proposal. lr, wd, m and R denote learning rate, weight decay, server momentum and residual-head indicator. Development selection is followed by evaluation at every prespecified heldout prefix; the full attempted-slot trace is retained separately.

Design	Learning rate	Weight decay	Server momentum	Residual head
D00	0.001	0.001	0	No
D01	0.0001	0.001	0	No
D02	0.01	0.001	0	No
D03	0.001	0	0	No
D04	0.001	0.01	0	No
D05	0.001	0.001	0.9	No
D06	0.001	0.001	0	Yes
D07	0.001	0.001	0.9	Yes
D08	0.0001	0.001	0	Yes
D09	0.0001	0.001	0.9	Yes
D10	0.01	0.01	0	No
D11	0.001	0.01	0	Yes

Table B.11: Twelve executable configurations for the short6 protocol. Every design has proximal coefficient zero. A trace design ID uniquely identifies these parameter changes; the task-specific prediction interface remains fixed.

Design	Learning rate	Weight decay	Server momentum	Residual head
L0W0M0R0	0.0001	0	0	No
L0W0M0R1	0.0001	0	0	Yes
L0W0M1R0	0.0001	0	0.9	No
L0W0M1R1	0.0001	0	0.9	Yes
L0W1M0R0	0.0001	0.001	0	No
L0W1M0R1	0.0001	0.001	0	Yes
L0W1M1R0	0.0001	0.001	0.9	No
L0W1M1R1	0.0001	0.001	0.9	Yes
L0W2M0R0	0.0001	0.01	0	No
L0W2M0R1	0.0001	0.01	0	Yes
L0W2M1R0	0.0001	0.01	0.9	No
L0W2M1R1	0.0001	0.01	0.9	Yes
L1W0M0R0	0.001	0	0	No
L1W0M0R1	0.001	0	0	Yes
L1W0M1R0	0.001	0	0.9	No
L1W0M1R1	0.001	0	0.9	Yes
L1W1M0R0	0.001	0.001	0	No
L1W1M0R1	0.001	0.001	0	Yes
L1W1M1R0	0.001	0.001	0.9	No
L1W1M1R1	0.001	0.001	0.9	Yes
L1W2M0R0	0.001	0.01	0	No
L1W2M0R1	0.001	0.01	0	Yes
L1W2M1R0	0.001	0.01	0.9	No
L1W2M1R1	0.001	0.01	0.9	Yes
L2W0M0R0	0.01	0	0	No
L2W0M0R1	0.01	0	0	Yes
L2W0M1R0	0.01	0	0.9	No
L2W0M1R1	0.01	0	0.9	Yes
L2W1M0R0	0.01	0.001	0	No
L2W1M0R1	0.01	0.001	0	Yes
L2W1M1R0	0.01	0.001	0.9	No
L2W1M1R1	0.01	0.001	0.9	Yes
L2W2M0R0	0.01	0.01	0	No
L2W2M0R1	0.01	0.01	0	Yes
L2W2M1R0	0.01	0.01	0.9	No
L2W2M1R1	0.01	0.01	0.9	Yes

Table B.12: Thirty-six executable configurations for the long24 protocol. Every design has proximal coefficient zero. A trace design ID uniquely identifies these parameter changes; the task-specific prediction interface remains fixed.

Table B.13: Complete proposal trace for short6. Each token is slot:design:decision, with A=retained, R=not retained, I=invalid. Design configurations are decoded in Table B.11. All proposed slots, including failures, are retained. Full configuration deltas and development scores are in the accompanying snapshot and TSV.

Task	Proposer	Mode	Slot:design:decision
DTI	Luna	Direct	1:D06:R; 2:D05:R; 3:D04:R; 4:D07:R; 5:D03:R; 6:D11:A
DTI	Luna	Loop	1:D06:R; 2:D08:R; 3:D05:R; 4:D04:R; 5:D03:R; 6:D01:R
DTI	Qwen	Direct	1:D06:R; 2:D07:R; 3:D08:R; 4:D09:R; 5:D04:R; 6:D05:R
DTI	Qwen	Loop	1:D06:R; 2:D07:R; 3:D08:R; 4:D09:R; 5:D04:R; 6:D05:R
PTPC	Luna	Direct	1:D06:R; 2:D05:R; 3:D03:R; 4:D07:R; 5:D04:R; 6:D11:R
PTPC	Luna	Loop	1:D06:R; 2:D08:R; 3:D05:R; 4:D04:R; 5:D11:R; 6:D03:R
PTPC	Qwen	Direct	1:D06:R; 2:D07:R; 3:D08:R; 4:D09:R; 5:D04:R; 6:D05:R
PTPC	Qwen	Loop	1:D06:R; 2:D10:R; 3:D07:R; 4:D08:R; 5:--I; 6:D09:R
VCC	Luna	Direct	1:D06:R; 2:D07:R; 3:D03:A; 4:D04:R; 5:D05:R; 6:D11:R
VCC	Luna	Loop	1:D06:R; 2:D04:R; 3:D05:A; 4:D09:R; 5:D02:R; 6:D03:A
VCC	Qwen	Direct	1:D06:R; 2:D07:R; 3:D08:R; 4:D09:R; 5:D04:R; 6:D11:R
VCC	Qwen	Loop	1:D06:R; 2:D07:R; 3:D10:R; 4:D08:R; 5:D04:R; 6:D09:R
Norman	Luna	Direct	1:D06:A; 2:D07:A; 3:D05:R; 4:D03:R; 5:D08:A; 6:D11:R
Norman	Luna	Loop	1:D06:A; 2:D11:R; 3:D05:R; 4:D04:A; 5:D01:A; 6:D09:R
Norman	Qwen	Direct	1:D06:A; 2:D07:A; 3:D08:A; 4:D09:R; 5:D04:R; 6:D11:R
Norman	Qwen	Loop	1:D06:A; 2:D07:A; 3:D11:R; 4:D08:A; 5:D10:R; 6:D05:R
Tahoe	Luna	Direct	1:D06:R; 2:D03:R; 3:D05:R; 4:D07:R; 5:D08:R; 6:D11:A
Tahoe	Luna	Loop	1:D06:R; 2:D05:R; 3:D04:A; 4:D11:R; 5:D10:R; 6:D03:R
Tahoe	Qwen	Direct	1:--I; 2:D06:R; 3:D07:R; 4:--I; 5:D08:R; 6:D09:R
Tahoe	Qwen	Loop	1:--I; 2:D06:R; 3:--I; 4:D07:R; 5:D09:R; 6:D05:R

Table B.14: Complete proposal trace for long24. Each token is slot:design:decision, with A=retained, R=not retained, I=invalid. Design configurations are decoded in Table B.12. All proposed slots, including failures, are retained. Full configuration deltas and development scores are in the accompanying snapshot and TSV.

Task	Proposer	Mode	Slot:design:decision
PTPC	Luna	Direct	1:L1W1M1R1:R; 2:L1W1M0R1:R; 3:L1W1M1R0:R; 4:L1W2M1R1:R; 5:L0W1M0R1:R; 6:L1W2M0R0:R; 7:L2W1M0R0:R; 8:L1W0M0R1:R; 9:L1W0M1R0:R; 10:L1W2M0R1:R; 11:L1W0M0R0:R; 12:L1W2M1R0:R; 13:L0W1M1R1:R; 14:L0W1M0R0:R; 15:L1W0M1R1:R; 16:L0W1M1R0:R; 17:L2W1M0R1:R; 18:L0W2M0R0:R; 19:L0W2M0R1:R; 20:L2W1M1R0:R; 21:L2W2M0R1:R; 22:L0W2M1R1:R; 23:L0W0M1R0:R; 24:L0W0M0R1:R
PTPC	Luna	Loop	1:L1W1M1R1:R; 2:L1W2M0R0:R; 3:L1W1M0R1:R; 4:L0W1M0R0:R; 5:L1W1M1R0:R; 6:L1W2M0R1:R; 7:L0W0M0R1:R; 8:L1W0M0R1:R; 9:L0W1M0R1:R; 10:L2W1M0R0:R; 11:L2W2M0R1:R; 12:L2W2M0R0:R; 13:L0W1M1R0:R; 14:L0W0M1R0:R; 15:L1W2M1R0:R; 16:L0W1M1R1:R; 17:L1W2M1R1:R; 18:L0W2M1R0:R; 19:L2W1M0R1:R; 20:L0W2M0R0:R; 21:L0W0M0R0:R; 22:L0W2M0R1:R; 23:L2W0M1R1:R; 24:L0W0M1R1:R

Task	Proposer	Mode	Slot:design:decision
PTPC	Qwen	Direct	1:L1W0M0R1:R; 2:L0W0M0R1:R; 3:L1W0M1R1:R; 4:L1W1M1R1:R; 5:L1W2M1R1:R; 6:L0W0M1R1:R; 7:L1W1M0R1:R; 8:L0W1M0R1:R; 9:L0W1M1R1:R; 10:L2W2M1R1:R; 11:L2W0M1R1:R; 12:--I; 13:L1W2M1R0:R; 14:L2W2M0R1:R; 15:--I; 16:L2W1M1R1:R; 17:--I; 18:L2W1M0R1:R; 19:--I; 20:L1W1M1R0:R; 21:--I; 22:L2W0M0R1:R; 23:L2W2M1R0:R; 24:L0W2M0R1:R
PTPC	Qwen	Loop	1:L1W0M0R1:R; 2:L1W0M0R0:R; 3:--I; 4:L0W0M0R1:R; 5:L0W0M1R1:R; 6:L0W1M1R1:R; 7:--I; 8:L0W1M0R1:R; 9:L1W1M1R1:R; 10:--I; 11:--I; 12:L1W0M1R1:R; 13:L1W1M0R1:R; 14:L2W2M1R1:R; 15:L1W2M1R1:R; 16:--I; 17:--I; 18:--I; 19:L0W2M0R1:R; 20:L1W1M1R0:R; 21:--I; 22:L1W2M0R1:R; 23:--I; 24:--I
VCC	Luna	Direct	1:L1W1M0R1:R; 2:L1W1M1R0:A; 3:L1W1M1R1:R; 4:L1W2M0R1:R; 5:L1W2M1R1:R; 6:L1W0M0R1:R; 7:L1W2M1R0:R; 8:L1W2M0R0:R; 9:L1W0M0R0:A; 10:L0W1M0R1:R; 11:L2W1M0R0:R; 12:L1W0M1R0:R; 13:L1W0M1R1:R; 14:L2W1M1R1:R; 15:L2W1M0R1:R; 16:L2W1M1R0:A; 17:L0W1M0R0:R; 18:L0W1M1R1:R; 19:L0W1M1R0:R; 20:L2W2M0R1:R; 21:L0W2M0R1:R; 22:L0W0M0R1:R; 23:L2W0M0R0:R; 24:L0W2M1R1:R
VCC	Luna	Loop	1:L1W1M0R1:R; 2:L2W2M0R0:R; 3:L1W1M1R0:A; 4:L0W1M1R0:R; 5:L1W1M1R1:R; 6:L1W0M1R0:R; 7:L1W2M1R0:R; 8:L0W0M1R1:R; 9:L2W1M1R0:A; 10:L2W2M1R0:A; 11:L2W2M1R1:R; 12:L1W2M1R1:R; 13:L1W2M0R0:R; 14:L1W0M0R0:R; 15:L1W0M1R1:R; 16:L2W2M0R1:R; 17:L1W2M0R1:R; 18:L0W1M0R0:R; 19:L1W0M0R1:R; 20:L2W1M0R0:R; 21:L2W1M0R1:R; 22:L0W1M0R1:R; 23:L2W1M1R1:R; 24:L0W0M0R0:R
VCC	Qwen	Direct	1:L1W1M1R1:R; 2:L1W1M0R1:R; 3:L0W0M0R1:R; 4:L0W0M1R1:R; 5:--I; 6:--I; 7:L0W1M0R1:R; 8:L1W2M1R1:R; 9:--I; 10:--I; 11:--I; 12:L0W1M1R0:R; 13:--I; 14:--I; 15:--I; 16:--I; 17:--I; 18:L2W2M1R1:A; 19:L1W0M1R1:R; 20:--I; 21:--I; 22:--I; 23:L0W1M1R1:R; 24:L1W2M0R1:R
VCC	Qwen	Loop	1:L1W1M1R1:R; 2:L0W0M1R1:R; 3:L1W0M0R1:R; 4:--I; 5:L0W0M0R1:R; 6:--I; 7:L0W1M1R1:R; 8:L0W1M0R1:R; 9:L1W1M0R1:R; 10:L1W2M1R1:R; 11:L1W2M1R0:R; 12:L0W2M0R1:R; 13:L1W1M1R0:A; 14:--I; 15:L1W0M1R1:R; 16:--I; 17:--I; 18:L2W0M1R1:R; 19:--I; 20:L1W2M0R1:R; 21:--I; 22:L2W1M1R1:R; 23:L2W2M1R1:A; 24:--I
Norman	Luna	Direct	1:L1W1M0R1:A; 2:L1W1M1R1:A; 3:L1W0M0R1:R; 4:L1W1M1R0:R; 5:L1W0M1R0:R; 6:L1W2M0R1:R; 7:L1W2M1R1:A; 8:L1W2M0R0:R; 9:L1W0M1R1:R; 10:L0W0M0R1:A; 11:L0W1M0R1:A; 12:L1W2M1R0:R; 13:L0W1M1R1:R; 14:L2W1M0R1:R; 15:L1W0M0R0:R; 16:L2W1M0R0:R; 17:L2W1M1R1:R; 18:L2W2M1R1:R; 19:L2W0M0R1:R; 20:L2W2M0R1:R; 21:L2W0M1R1:R; 22:L0W1M0R0:A; 23:L2W1M1R0:R; 24:L0W1M1R0:R
Norman	Luna	Loop	1:L1W1M0R1:A; 2:L0W1M0R1:A; 3:L0W1M1R1:R; 4:L0W1M0R0:A; 5:L0W1M1R0:R; 6:L0W2M0R0:R; 7:L0W0M0R0:A; 8:L0W2M0R1:R; 9:L1W0M0R0:R; 10:L0W0M0R1:R; 11:L1W0M0R1:R; 12:L0W0M1R0:R; 13:L1W1M1R0:R; 14:L1W2M0R0:R; 15:L0W2M1R0:R; 16:L0W2M1R1:R; 17:L0W0M1R1:R; 18:L2W1M0R0:R; 19:L1W1M1R1:R; 20:L1W2M0R1:R; 21:L1W2M1R0:R; 22:L2W2M0R0:R; 23:L1W0M1R0:R; 24:L1W0M1R1:R
Norman	Qwen	Direct	1:L1W1M0R1:A; 2:L0W0M0R1:A; 3:L0W1M1R1:R; 4:--I; 5:L1W1M1R1:R; 6:L0W0M1R1:R; 7:--I; 8:L2W2M1R1:R; 9:L0W1M0R1:A; 10:--I; 11:L1W2M0R1:R; 12:L1W2M1R1:R; 13:--I; 14:--I; 15:--I; 16:--I; 17:L2W2M0R1:R; 18:L2W1M0R1:R; 19:L2W0M1R1:R; 20:L2W1M1R1:R; 21:--I; 22:L0W2M0R1:A; 23:--I; 24:L1W0M0R1:R

Task	Proposer	Mode	Slot:design:decision
Norman	Qwen	Loop	1:L1W1M0R1:A; 2:-:I; 3:-:I; 4:L1W0M0R1:A; 5:L0W0M0R1:A; 6:L1W2M1R1:R; 7:-:I; 8:L0W0M1R1:R; 9:-:I; 10:-:I; 11:-:I; 12:L1W1M1R1:R; 13:-:I; 14:L0W1M1R1:R; 15:L1W2M0R0:R; 16:L1W1M1R0:R; 17:-:I; 18:-:I; 19:L0W1M0R1:A; 20:-:I; 21:L1W2M0R1:R; 22:-:I; 23:-:I; 24:L0W2M1R1:R
Tahoe	Luna	Direct	1:L1W1M0R1:R; 2:L1W1M1R1:R; 3:L1W1M1R0:R; 4:L1W2M0R1:A; 5:L1W2M1R1:A; 6:L1W0M0R1:R; 7:L1W0M0R0:R; 8:L1W2M1R0:R; 9:L1W2M0R0:R; 10:L1W0M1R1:R; 11:L0W1M0R0:R; 12:L0W0M0R1:R; 13:L1W0M1R0:R; 14:L0W1M0R1:R; 15:L0W1M1R1:R; 16:L2W1M0R0:R; 17:L0W0M1R1:R; 18:L2W1M1R0:R; 19:L0W1M1R0:R; 20:L2W1M0R1:R; 21:L2W1M1R1:R; 22:L0W2M0R1:R; 23:L2W2M1R1:R; 24:L0W2M1R1:R
Tahoe	Luna	Loop	1:L1W1M0R1:R; 2:L1W1M1R0:R; 3:L1W2M0R0:A; 4:L0W2M0R0:R; 5:L1W2M0R1:R; 6:L2W1M0R0:R; 7:L0W1M0R0:R; 8:L0W2M0R1:R; 9:L2W2M0R0:R; 10:L0W2M1R0:A; 11:L0W2M1R1:R; 12:L1W2M1R0:R; 13:L1W2M1R1:A; 14:L0W1M1R1:R; 15:L0W0M1R0:R; 16:L0W1M1R0:R; 17:L0W0M0R0:R; 18:L1W0M1R1:R; 19:L2W1M0R1:R; 20:L0W1M0R1:R; 21:L1W0M0R1:R; 22:L0W0M1R1:R; 23:L0W0M0R1:R; 24:L1W0M0R0:R
Tahoe	Qwen	Direct	1:L1W0M0R1:A; 2:L1W1M0R1:R; 3:-:I; 4:-:I; 5:L0W0M0R1:R; 6:-:I; 7:-:I; 8:L1W1M1R1:R; 9:L0W0M1R0:A; 10:L0W0M1R1:R; 11:-:I; 12:L0W1M0R1:R; 13:L1W2M1R0:R; 14:-:I; 15:L2W1M1R1:R; 16:L0W1M1R1:R; 17:L2W2M1R1:R; 18:L1W2M1R1:A; 19:-:I; 20:-:I; 21:L1W2M0R1:R; 22:L1W0M1R1:R; 23:-:I; 24:-:I
Tahoe	Qwen	Loop	1:L1W0M0R1:A; 2:-:I; 3:-:I; 4:-:I; 5:L1W1M1R1:R; 6:-:I; 7:L1W1M0R1:R; 8:L1W0M1R1:A; 9:L0W0M1R1:R; 10:L0W0M0R1:R; 11:-:I; 12:-:I; 13:-:I; 14:L1W2M1R1:A; 15:L2W2M1R0:R; 16:L0W1M1R1:R; 17:-:I; 18:L0W1M0R1:R; 19:L0W1M1R0:R; 20:-:I; 21:L1W2M0R1:R; 22:L2W1M1R1:R; 23:L2W2M1R1:R; 24:L0W2M0R1:R